

# Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Progress update — the modelling half, and an honest *negative* result

Kostiantyn Boiar

MSc Dissertation ■ University of Glasgow

Supervisor: Dr Marwa Mahmoud

July 23, 2026

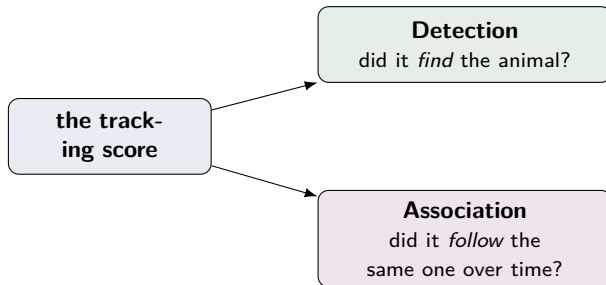
## The result, in one sentence

We can **measure** SAM 3's reliability — but we **cannot predict it in advance** from label-free “distance” signals, beyond a small and obvious **animal-size** effect.

- This is a **clean, rigorous negative result** — and it is the contribution.
- It is *not* a weak-test artefact: the same test *does* pick up the size effect (shown later).
- Everything below is built, tested, and validated out-of-sample with honest statistics.

# The question

Point SAM 3 at a video, type “impala”, and it finds and follows the animal — **with no training on that species**. Can we tell, *before running it*, whether it will work for a **new animal** in a **new place** — **without labelling anything first**?



We split the question in two — that split is the intellectual core of the project.

# What we've done

1. ✓ **Measured the model.** Ran frozen SAM 3 over the test videos and scored it with the *official* evaluator — a reliability number for each (species, place, time) group.
2. ✓ **Built four “distance” signals** — how *different* a new species/place is from the familiar training data: the animal (tree-of-life + appearance), the scene, and the time gap.
3. ✓ **Fitted a proper model** (score = function of the distances) and tested it **honestly out-of-sample**: hide whole species and whole places, predict them, compare to the truth — with conservative, group-aware error bars.
4. ✓ **Tried a second idea** when the first failed — maybe it is how *hard* the footage is (dark/night, cluttered, small animal), not how *novel* it is.
5. ✓ **Checked a side question** — when the animal is *absent*, does the model hallucinate one?

## What the pipeline sees — a real frame



frame + ground-truth mask



**visual:** the animal alone



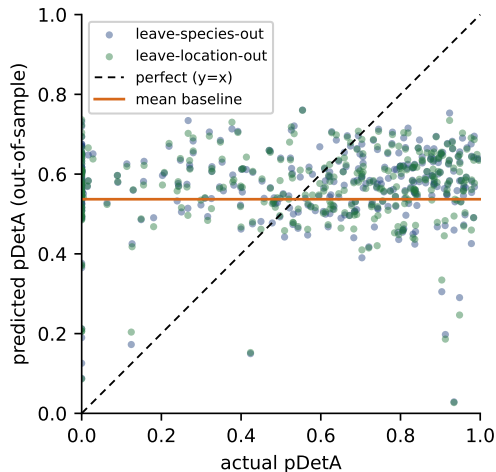
**environment:** animal erased

The distances are built from label-free crops of a real camera-trap frame (a collared peccary): the animal's appearance (*visual*), the scene with the animal erased (*environment*), plus its place on the tree of life and its timestamp. **No SAM 3 is used to build them.**

# The main result: the distances predict *nothing*

| Predict from      | error | beats "guess"? |
|-------------------|-------|----------------|
| guess the average | 0.30  | —              |
| + four distances  | 0.30  | — <b>no</b>    |
| + animal size     | 0.28  | ✓ a little     |

out-of-sample error, held-out species / places  
(lower is better)



predictions collapse onto the *mean*, not the diagonal

## The one “signal” was a trick of animal size

- One distance (**visual**) looked significant — but with the *wrong* sign, and it was a **confound**: “visually distinctive species do better” really means “**bigger animals are easier to spot.**”
- Add an animal-size control and the effect **vanishes** (coefficient  $+0.37 \rightarrow +0.22$ , now overlapping zero); size itself is the real driver.

### The “difficulty” idea collapsed the same way

Dark / cluttered footage *looked* predictive — but only because it tracks animal size. On its own it predicts nothing (a strong-looking  $r = -0.38$  across places shrinks to  $-0.13$  per clip: a classic grouping illusion). **The difficulty pivot is also a null.**

## Peeling it apart: *only size* survives

| Predict detection from... | Beats guessing? | Gain          |
|---------------------------|-----------------|---------------|
| clip length only          | — no            | $\approx 0$   |
| <b>animal size</b> only   | ✓ <b>yes</b>    | +0.017        |
| low-light only (no size)  | — no            | +0.004 (n.s.) |
| low-light + size          | ✓ yes (= size)  | +0.020        |

- Size **alone** carries the whole gain; low-light **alone** does not validate.
- So the *only* label-free property that predicts detection is **object size** — small, obvious, and the very thing we introduced as a **nuisance control**.



## “Following” barely differs from “finding”

- Almost every clip has a **single animal**, so the two scores are nearly identical — they agree about **94%** of the time.
- There is very little separate “following” signal for *any* feature to predict.
- So the association half is honestly a **detection story**; we report its null with those numbers as the explanation, not as a separate failed search.

The detection-vs-association split was the right question to ask — the honest finding is that, on this data, association carries almost no distinct signal.

# Is “we found nothing” trustworthy?

## The obvious worry

*“Maybe the test just isn’t sensitive enough.”*

- The **same test**, on the **same data**, *does* detect the animal-size effect out-of-sample.
- A test that **fires for size** but **stays flat for the distances** is genuinely finding *no signal* — not failing to look.

**This is the key defence of the negative result.**

# The null survives every check

**Drop any one distance** — the fit and the (null) out-of-sample gain barely move. Only **adding animal size** changes anything.

| Model              | pseudo- $R^2$ | beats guess? |
|--------------------|---------------|--------------|
| all four distances | 0.055         | — no         |
| drop any one       | 0.04–0.06     | — no         |
| + animal size      | 0.119         | ✓ yes        |

- The four distances together explain only  $\approx 3\%$  of the variance.
- No factor dominates, and every collinearity score is  $\approx 1$  — so it is **not** collinearity hiding a real effect.
- The null is not an artefact of one factor or one modelling choice.

## Two other things we checked

### Hallucinations (animal absent)

SAM 3 wrongly returns an animal about **10%** of the time. More distinctive species hallucinate *less* (a real, size-independent correlation) — but out-of-sample it **just misses** significance ( $p = 0.053$ ). A genuine correlation, **not** a validated predictor.

### The model's own confidence (explored, then set aside)

If we let SAM 3 *run first* and read its own confidence, that *does* predict its accuracy. We deliberately keep it out of the contribution: (1) it needs running the model, defeating the “predict *before* running” goal; and (2) it is near-circular (a known result). We use it only as an internal check that the pipeline can detect a strong signal.